

Task 1 — Geometrically Weighted Terms in ERGMs (Glasgow Wave 1)

All computations use the Wave 1 directed friendship network (`Glasgow/f1.csv`) with $n = 129$ students (diagonal fixed to 0). The observed network has $m = 448$ directed edges (density $m / (n(n-1)) = 0.02713$), and the maximum observed out-degree is 6.

To reproduce plots/tables in this write-up, run:

```
Rscript Task1.R
```

```
## Observed density: 0.02713
## M1 theta_edges: -3.58
## M2 theta_edges (theta_gwd=-2, alpha=0.3): -3.368
## M2 theta_edges (theta_gwd=+2, alpha=0.3): -3.702
## Plots written to task1_figures/
## Tables written to task1_results/
```

This creates figures in `task1_figures/` and intermediate tables in `task1_results/`.

(1) Change statistic of `gwodegree`

(1.1) Derive the change statistic

The statistic is

$$s_{\text{GWD}}(x, \alpha) = e^{\alpha} \sum_{k=1}^n (1 - (1 - e^{-\alpha})^k) D_k^{\text{out}}(x).$$

Consider toggling on one directed edge from a sender whose current out-degree is k . Then exactly one node changes out-degree from k to $k + 1$, so: $- D_k^{\text{out}}$ decreases by 1 and $- D_{k+1}^{\text{out}}$ increases by 1.

Therefore the change statistic is

$$\begin{aligned}
\delta_{\text{GWD}}(k, \alpha) &= e^\alpha \left[(1 - (1 - e^{-\alpha})^{k+1}) - (1 - (1 - e^{-\alpha})^k) \right] \\
&= e^\alpha \left[(1 - e^{-\alpha})^k - (1 - e^{-\alpha})^{k+1} \right] \\
&= e^\alpha (1 - e^{-\alpha})^k \left[1 - (1 - e^{-\alpha}) \right] \\
&= e^\alpha (1 - e^{-\alpha})^k e^{-\alpha} \\
&= (1 - e^{-\alpha})^k.
\end{aligned}$$

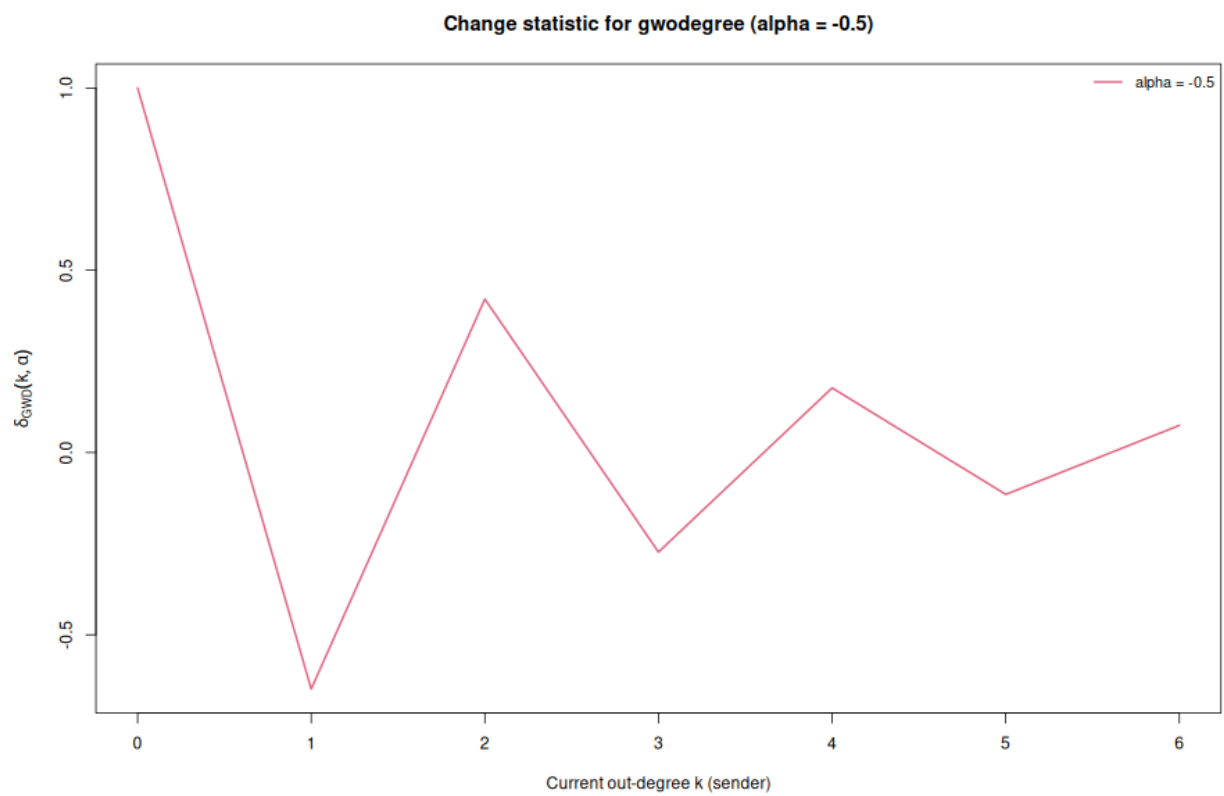
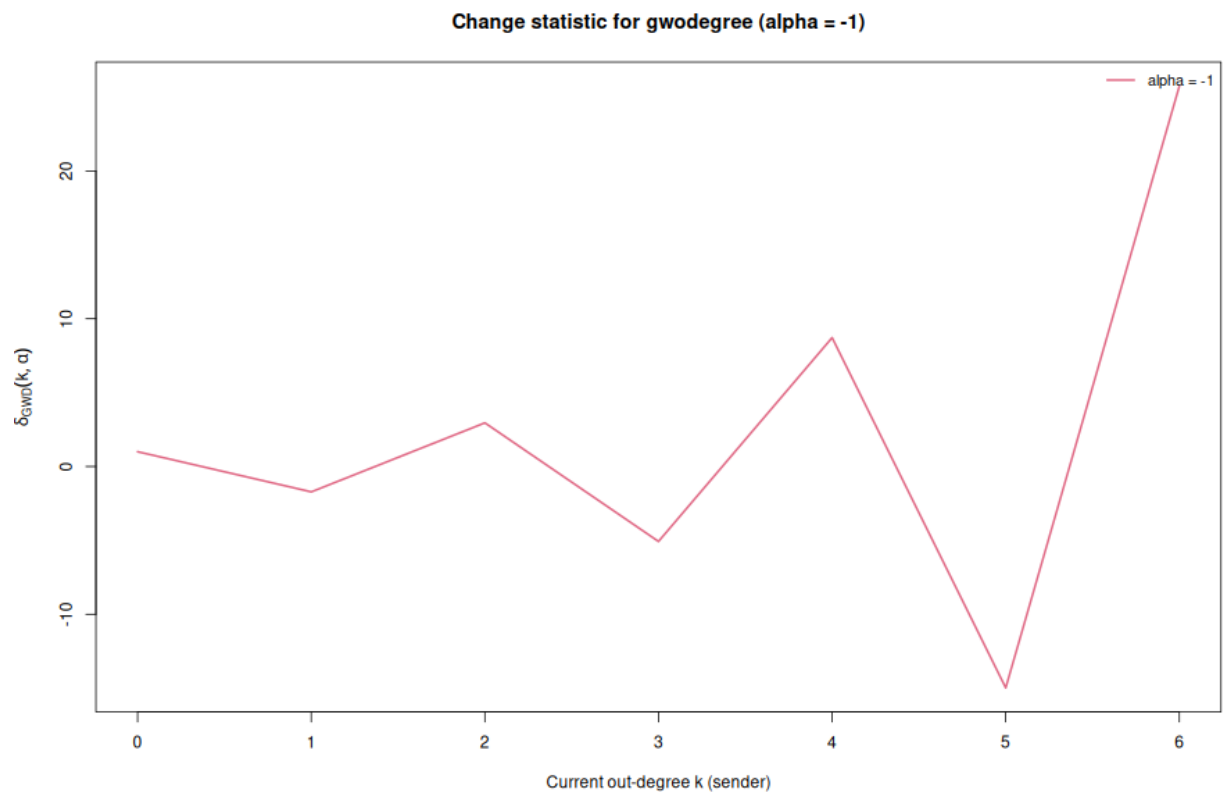
So,

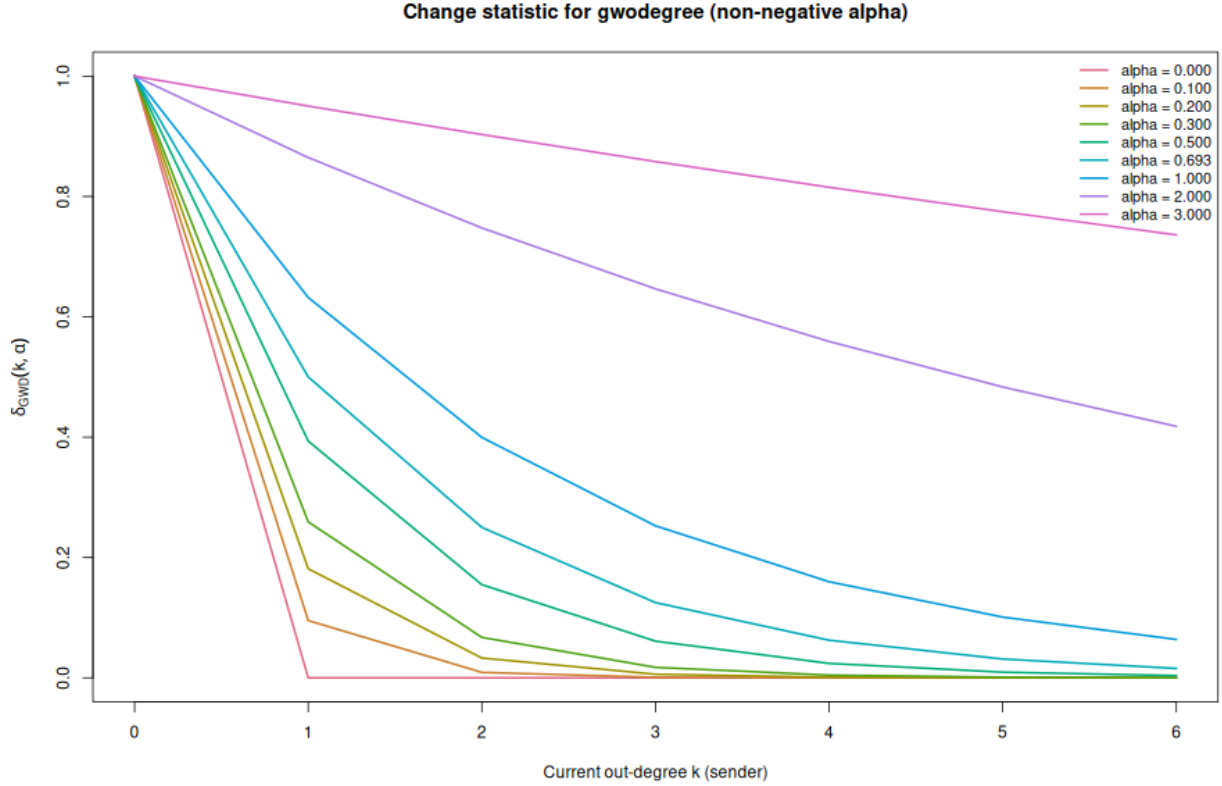
$$\delta_{\text{GWD}}(k, \alpha) = (1 - e^{-\alpha})^k \quad \text{for } k \geq 0.$$

(1.2) Plots of $\delta_{\text{GWD}}(k, \alpha)$ vs. k

Plots were produced for $k = 0, \dots, 6$ (the maximum observed out-degree in Glasgow Wave 1).

- Negative decay values:
 - `task1_figures/gwodegree_change_alpha_-1.png`
 - `task1_figures/gwodegree_change_alpha_-0.5.png`
- Non-negative decay values:
 - `task1_figures/gwodegree_change_alpha_nonneg.png`





(1.3) Main takeaways

From the analytic expression $\delta_{\text{GWD}}(k, \alpha) = (1 - e^{-\alpha})^k$, the decay parameter α directly controls how the marginal contribution of an additional out-tie changes with the sender's current out-degree.

For non-negative α we have $0 \leq 1 - e^{-\alpha} < 1$, so $\delta_{\text{GWD}}(k, \alpha)$ is non-negative and decreases exponentially in k . This means additional outgoing ties have the largest marginal contribution when the sender currently has low out-degree, and the marginal contribution becomes small for already-active senders. The strength of this “diminishing returns” behaviour is governed by α : when α is small and positive, $1 - e^{-\alpha}$ is close to 0, so the marginal contribution drops to near-zero after only a few outgoing ties. In the limiting case $\alpha = 0$, the first outgoing tie contributes 1 (at $k = 0$) and further outgoing ties contribute 0, so the statistic essentially counts how many actors have at least one outgoing tie. When α is large, $1 - e^{-\alpha}$ approaches 1 and $\delta_{\text{GWD}}(k, \alpha)$ stays close to 1 even for larger k . In that regime **gwodegree** becomes increasingly similar to an edges-like term (each additional edge changes the statistic by almost the same amount), so the term loses its ability to meaningfully distinguish between moderate and high out-degrees.

For negative α , $1 - e^{-\alpha} < 0$ and the change statistic alternates sign with k . For $\alpha = -0.5$, $|1 - e^{-\alpha}| < 1$ so the magnitude still decays with k (but oscillates in sign). For $\alpha = -1$, $|1 - e^{-\alpha}| > 1$ so the magnitude grows rapidly with k , making the marginal effect extremely sensitive to already-high out-degree. This behaviour is typically undesirable in ERGMs because it can lead to unstable contributions and counter-intuitive incentives for adding/removing ties depending on the parity of k .

(1.4) Choosing α for an “activity-among-the-active” effect

An “activity-among-the-active” effect can be understood in two ways. If we mean that an additional outgoing tie should have a larger marginal contribution when the sender is already active, then the usual non-negative `gdegree` term cannot really do this. The change statistic is

$$\delta_{\text{GWD}}(k, \alpha) = (1 - e^{-\alpha})^k.$$

For $\alpha \geq 0$, this is non-increasing in k . Thus the marginal contribution of an additional outgoing tie is largest for low out-degree senders and becomes smaller as the sender’s current out-degree grows.

However, if the goal is that the statistic remains sensitive to differences among already-active senders, then α should be chosen relatively large and positive. A larger α makes $1 - e^{-\alpha}$ closer to 1, so $\delta_{\text{GWD}}(k, \alpha)$ decays more slowly with k . This means that adding a 4th, 5th, or 6th outgoing tie can still make a meaningful contribution to the statistic. In contrast, with small positive decay, the change statistic drops close to zero after only a few outgoing ties, so the term mostly distinguishes inactive or weakly active actors from others.

There is a trade-off. If α is too large, $\delta_{\text{GWD}}(k, \alpha) \approx 1$ for all relevant k , and `gdegree` becomes close to an edges-like or linear out-degree term. If α is too small, the term ignores variation among high-out-degree actors. Negative α can make the absolute change grow with k , but the alternating sign makes it hard to interpret and potentially unstable. Therefore, a moderate-to-large positive decay is the most appropriate choice.

(2) Effect of the `gdegree` coefficient on the out-degree distribution

(2.1) Models

- M1 (reference): edges-only ERGM. This is equivalent to an i.i.d. Bernoulli model for directed edges with p = observed density, and $\hat{\theta}_{\text{edges}} = \text{logit}(p)$.
- M2: edges + `gdegree` with fixed $\alpha = 0.3$ and fixed $\theta_{\text{GWD}} \in \{-2, +2\}$. For each fixed θ_{GWD} , θ_{edges} is chosen so that the expected density matches the observed density (simulation-based matching).

The fitted parameters (reproducible with the provided script) are:

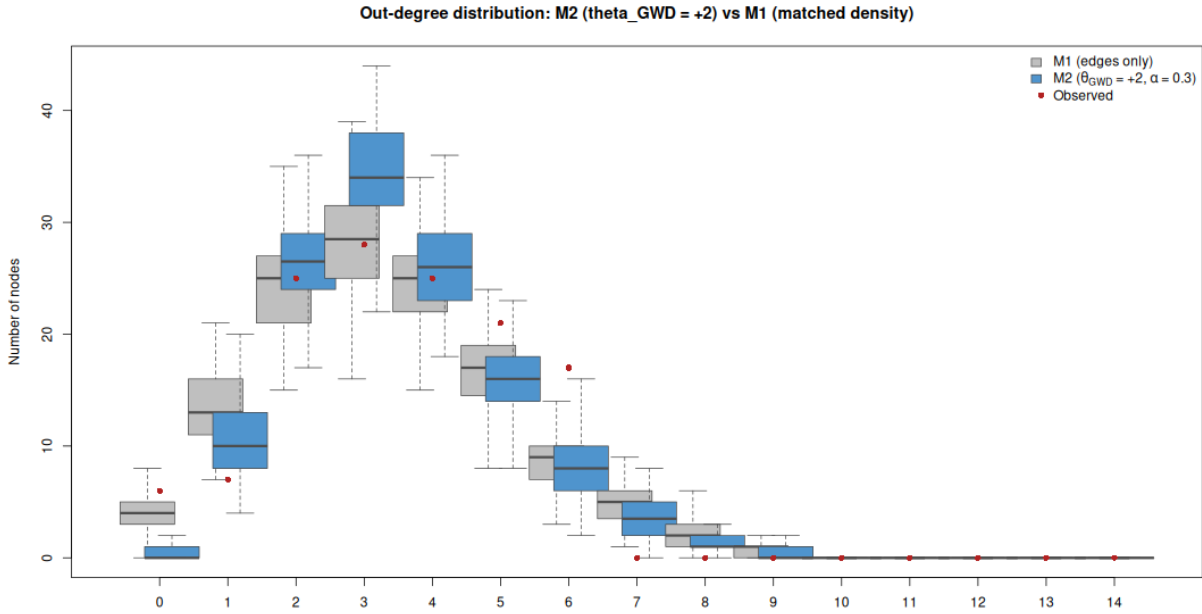
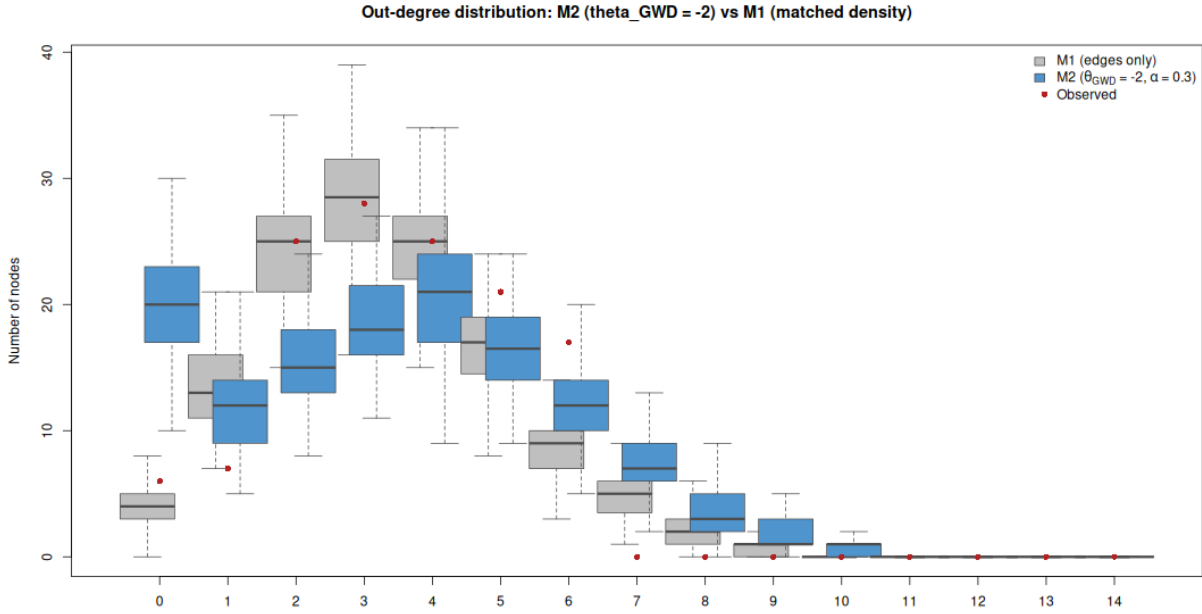
Model	θ_{edges}	θ_{GWD}	α
M1	-3.580	0	–
M2	-3.368	-2	0.3
M2	-3.702	+2	0.3

The same values are written to `task1_results/model_parameters.csv`.

(2.2) Simulation and box plots

For each of the three models, 100 networks were simulated and the node count at each out-degree was computed. For M2, results are plotted against M1 (reference), and the observed Wave 1 counts are shown as red points.

- task1_figures/outdegree_box_M2_gwd_-2.png
- task1_figures/outdegree_box_M2_gwd_+2.png



(2.3) Effects of θ_{GWD} on the out-degree distribution

Holding density fixed isolates how θ_{GWD} reshapes who sends ties, rather than how many ties exist overall. Relative to M1 (a random graph with the same density), changing θ_{GWD} primarily changes the heterogeneity of the out-degree distribution.

With $\theta_{\text{GWD}} = +2$ (and $\alpha = 0.3$), adding an outgoing tie yields a larger positive contribution when the sender has low out-degree (because $\delta_{\text{GWD}}(k, \alpha)$ is largest at small k). This tends to distribute outgoing nominations more evenly across students: compared to M1, simulations show fewer extreme out-degree values and more mass around moderate degrees. In other words, positive θ_{GWD} with a fixed positive decay acts like a “regularizer” on activity, discouraging highly active senders because the marginal gain from additional outgoing ties quickly becomes small once k is already moderate.

With $\theta_{\text{GWD}} = -2$, the same diminishing change statistic implies that adding early outgoing ties is penalized strongly (large negative change in log-odds at low k), while adding ties for already-active senders is penalized relatively little (small magnitude at high k). This creates a cumulative advantage process: once a sender has accumulated some outgoing ties, additional ties become comparatively easier to add. As a result, the out-degree distribution becomes more skewed than under M1: more students have very low out-degree, while a smaller set of students becomes highly active and concentrates outgoing nominations.

Overall, θ_{GWD} tunes the degree heterogeneity around the same mean density: positive values push toward more homogeneous activity and negative values push toward more heterogeneous activity, relative to the edges-only random graph baseline.